

Li Arditi

liarditi@gmail.com | [linkedin.com/in/li-arditi](https://www.linkedin.com/in/li-arditi) | [Engineering Portfolio](#)

PROJECT EXPERIENCE

Orthopedic Surgery Product Design & Development

Boston, MA

Master's Capstone Project | Team of 4

September 2025 - May 2026

- Collaborated with diverse clinical stakeholders during 50+ hours of OR observations to analyze clinical workflows, identify usability risks, and inform design inputs for unmet needs
- Designed, prototyped, tested, and maintained Design History File for development of novel skeletal traction device, aligning with regulated medical device development standards and design controls throughout full product lifecycle

Single-Serving Coffee Press Product Realization

Boston, MA

Mechanical Engineering Course Project | Team of 4

January 2026 - May 2026

- Designed, prototyped, and documented product development from proof-of-concept to mass production plan
- Established and documented standards, workflows, and automations in enterprise data management (EDM) plan and Standard Operating Procedures (SOPs)
- Machined parts with wire EDM, manual mill, lathe, water jet, TIG welding, and sewing machine

Endoscope Stabilization Sleeve Mock Grant Application

Boston, MA

Medical Robotics Course Project | Team of 5

January 2025 - May 2025

- Designed and prototyped a robotic endoscope stabilization system using pneumatic actuation, pressure sensing, and Arduino-based control and software components
- Prepared grant application materials in alignment with NIH guidelines, emphasizing technical performance, clinical significance, and innovation

PROFESSIONAL EXPERIENCE

Software-Hardware Integration Engineer

Boston, MA

Cognitive Neuroimaging Center at Boston University

May 2025 - September 2025

- Implemented new Python- and C#-based workflow to automate collection and synchronization of multimodal clinical data with millisecond precision across multiple devices, aligning with neuroscience and human research study quality requirements
- Documented 30-page technical report of research findings, validation and verification studies, system architecture and integrations, lessons learned, and next steps to support traceability, knowledge transfer, and future development

Virtual Design and Construction (VDC) Engineer

New York, NY

Turner Construction Company

October 2023 - February 2024

- Coordinated cross-functional team of 5-10 subcontractors to manage 3D models, collaborating with engineers and reviewing model data for compliance with specifications, identifying risks, trends, and discrepancies requiring escalation
- Presented design-related issues to internal teams and external clients, facilitating dialogue to drive collaborative problem-solving and ensure alignment with stakeholder expectations and industry regulations

Full-Stack Software Engineer

New York, NY

Turner Construction Company

March 2021 - June 2024

- Managed and collaborated with team members and end users to design, develop, and deploy full-stack custom web apps for real-time KPI dashboards to support management decisions and continuous improvement initiatives
- Recognized with 2022 Staff Award for outstanding contribution towards improvement of overall quality and performance of company and its service to clients and the industry

EDUCATION

Boston University

Boston, MA

Master of Engineering (MEng) in Biomedical Engineering, GPA: 3.96/4.00

May 2026

Relevant Courses: Biomedical Measurements; Medical Robotics; Advanced Biomedical Design and Development (Fall '25 - Sp '26); Feedback Control Theory for Biological Systems; Programming Fundamentals for BME Data Analysis; Biomedical Data Science & Machine Learning; Product Realization

The University of Chicago

Chicago, IL

Bachelor of Arts in Neuroscience, Minor in Computer Science, GPA: 3.56/4.00

June 2021

RELEVANT SKILLS

Programming: Python (NumPy, Pandas, SciPy), MATLAB, C#, SQL, JS, React, Git, CAD, Arduino, Raspberry Pi, PHP, HTML

Quality & Design: Human Factors Analysis, Requirement Traceability Matrix (RTM), Design for Manufacturability (DFM),

Validation and Verification Testing, Failure Modes and Effects Analysis (FMEA), Enterprise Data Management (EDM)

Standards and Regulations: OR Protocols, Human-Subject Research, FDA regulations and ISO standards, Machine Shop and Manufacturing

EXTRACURRICULAR AND LEADERSHIP ACTIVITIES

Turner Pride Network, Co-lead, LGBTQ+ Employee Resource Group

New York, NY | August 2022 - May 2024

- Promoted LGBTQ+ visibility, inclusion, and diversity throughout company and general construction industry with educational and social events, presentations, and panels